12.15 Extinction or Explosion

WORKSHEET

The question

STEP II 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If there are two particles at one stage, what is the distribution of the number at the next stage?

STAGE 1 Two, then two

Covers: the joint probability

Ask yourself What is $\mathrm{P}(X_1 = 2)$? If there are two particles, what is the distribution of the number in the next generation?

STAGE 2 Reasoning backwards

Covers: the conditional probability

Pause and try Find $\mathrm{P}(X_2 = 2)$ by splitting by the value of X_1. Then use the definition of conditional probability, and set the answer equal to $\frac{9}{25}$.

STAGE 3 The expected population

Covers: the expectation, and the limits

Ask yourself If there are k particles in one generation, what is the expected number in the next? Average over k, and solve the recurrence for $\mathrm{E}(X_n)$.

What's really going on

The probability of extinction

For $p = \frac{3}{4}$, the expected population grows without limit. Does that mean the population survives? Not necessarily: if the first particle has no offspring (probability $q^2 = \frac{1}{16}$), the line ends at once. Let e be the probability of eventual extinction. The first particle has k offspring, and each of their lines dies out independently with probability e, so

$$e = q^2 + 2pq\,e + p^2e^2 = (q + pe)^2 = \mathrm{G}(e),$$

where $\mathrm{G}(s) = (q + ps)^2$ is the *probability generating function* of the offspring number. The extinction probability is a fixed point of G: in fact the *smallest* non-negative one. For $p = \frac{3}{4}$, $(1 + 3e)^2 = 16e$ gives $9e^2 - 10e + 1 = 0$, so $e = \frac{1}{9}$ or 1, and the extinction probability is $\frac{1}{9}$. For $p = \frac{1}{4}$, the only fixed point in $[0, 1]$ is $e = 1$: extinction is certain

(Figure 12.15.1).

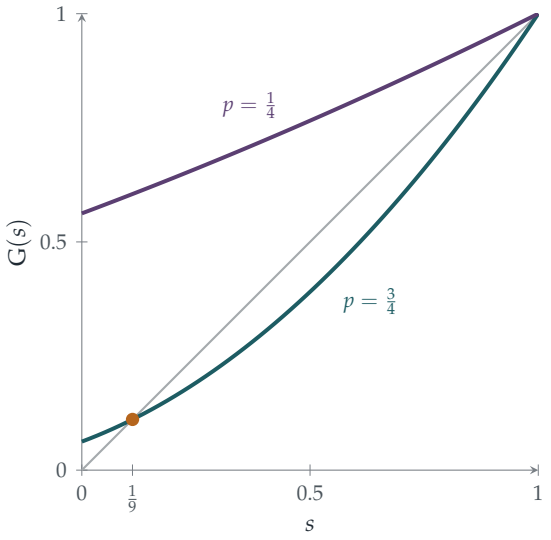


Figure 12.15.1 The offspring generating function $G(s) = (q + ps)^2$ and the line $y = s$. The probability of extinction is the smallest point where they meet. For $p = \frac{1}{4}$ (plum, mean $\frac{1}{2}$) that is $s = 1$: extinction is certain. For $p = \frac{3}{4}$ (teal, mean $\frac{3}{2}$) it is $s = \frac{1}{9}$ (amber): the population survives for ever with probability $\frac{8}{9}$.

The critical threshold

In general, a branching process dies out with certainty if the mean number of offspring is less than 1, and also if it equals 1, except in the trivial case where every individual has exactly one offspring (when the family line simply continues for ever, one at a time); it survives with positive probability if the mean is greater than 1. The mean offspring number is the *reproduction number* R of epidemiology: an epidemic with $R > 1$ may take off, but even then an early outbreak can fizzle out by chance, with probability given by exactly the fixed-point equation above. In nuclear physics, a chain reaction is *critical* when each fission causes on average exactly one more, and the art of running a reactor is to keep it there.

Galton, Watson and Bienaymé

Galton's question was answered, partly incorrectly, by the Reverend Henry William Watson in 1874: he found the fixed-point equation but concluded that every family name dies out eventually, missing the second root. The correct analysis had in fact been given in 1845 by the French statistician Irénée-Jules Bienaymé, whose work was forgotten until the 1970s. The model is now called the *Galton–Watson process* (sometimes Bienaymé–Galton–Watson), and is a central example in every course on random processes.

Try this next

1. **Extinction in general.** For offspring distribution $\mathrm{B}(2, p)$, show that the extinction probability is 1 if $p \le \frac{1}{2}$ and $\left(\frac{q}{p}\right)^2$ if $p > \frac{1}{2}$.

2. **The variance.** Show that $\mathrm{Var}(X_n)$ satisfies $\mathrm{Var}(X_n) = (2p)^2 \mathrm{Var}(X_{n-1}) + 2pq\,\mathrm{E}(X_{n-1})$, and find it for $p = \frac{1}{2}$.

3. **Extinction by generation n.** Let $e_n = \mathrm{P}(X_n = 0)$. Show that $e_n = \mathrm{G}(e_{n-1})$, and compute e_1, e_2, e_3 for $p = \frac{3}{4}$. Do they approach $\frac{1}{9}$?

4. **Surnames.** If each man has 0, 1, 2 or 3 sons with probabilities $\frac{1}{4}$ each, what is the probability that his surname eventually dies out?

Further reading

- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the section on branching processes.
- **Adam Kucharski, *The Rules of Contagion*** (Profile). Reproduction numbers and outbreaks, for a general reader.
- **David Kendall, "Branching processes since 1873"**, *Journal of the London Mathematical Society* 41 (1966). The history, including Bienaymé.

The full guided solution is in Section 12.15:
https://beyondstep.keithmcnulty.org/ch12/branching.html